

THERMODYNAMICS

✚ Important formulas and concepts directly from the NCERT-

- “Thermodynamics” is the branch of physics that deals with the concepts of heat and temp. and the interconversion of heat and other forms of energy.
- Thermodynamics is a microscopic science and it deals with bulk systems.

❖ Thermal Equilibrium-

- Thermodynamic equilibrium- a state with fixed values of pressure , volume , temperature , mass and composition that do not changes with time.
- A system is in the state of equilibrium or not depends on the surroundings and the nature of wall that separates the system from the surroundings.
- **In thermal equilibrium, Temperature of two systems are Equal.**

❖ Zeroth Law Of Thermodynamics-

- Acc. to this law, “two systems in thermal equilibrium with a third system separately are in thermal equilibrium with each other”.

❖ Heat , Internal energy and Work-

- Internal energy is simply the sum of kinetic energy and potential energy of the molecules.
- Kinetic energy of the system as a whole is not relevant .
- Thus, Internal energy is the energy associated with the random motion of molecules of system.
- Internal energy depends only on the state of the system , not on how the state was achieved.it is a ‘state variable’.
- Pressure, Volume, Temperature and Internal energy are thermodynamic state variables of the system.
- Heat and work are two different modes of altering the state of a thermodynamic system and changing its internal energy.

❖ First Law Of Thermodynamics-

- Based on the principle of “**conservation of energy**”
- Acc. to this law,

If ΔQ = heat supplied

ΔW = work done

ΔU = change in internal energy

Then, $\Delta Q = \Delta U + \Delta W$

- ΔU depends only on the initial and final states and not on the path taken.
- ΔQ and ΔW depend on the path taken .
- The combination ($\Delta Q - \Delta W$) is path independent.
- Work done against a constant pressure P ,

$$W = P\Delta V$$

First Law,

$$\Delta Q = \Delta U + P\Delta V$$

❖ Specific Heat Capacity-

- Heat required to increase temp. of unit mass by unit.

$$S = \Delta Q / m\Delta T$$

- Constant characteristic of the substance.
- Independent of amount of substance
- Depends on the nature of the substance and its temperature.
- J/kg K
- **Molar specific heat capacity** – heat capacity per mole.

$$C = \Delta Q / m\Delta T$$

- Independent of amount of substance.
- Depends on nature , temp. and conditions under which heat is supplied.
- Unit – J/mol k

- An oscillator in **one dimension** has average energy of $K_B T$.
- **In three dimensions**, the average energy is $3K_B T$.
- For a mole of solid , the total energy is

$$U = 3K_B T \times N_A = 3RT$$

- For a solid, ΔV is negligible
Therefore, from First Law,

$$\Delta Q = \Delta U$$

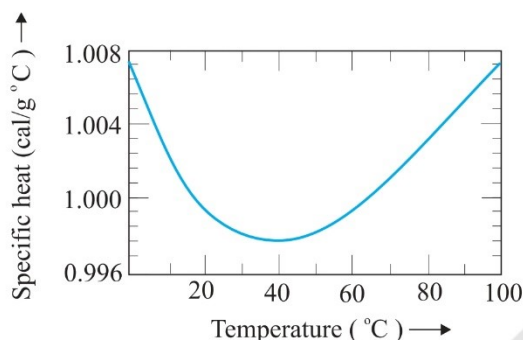
Heat capacity,

$$H = \Delta Q / \Delta T = \Delta U / \Delta T = 3R$$

Predicted value $3R$ at ordinary temp. (carbon is an exception)

○ Specific Heat Capacity of Water-

- one calorie is defined as the amount of heat required to raise temp of 1 gm of water by 1°C .
- specific heat of water varies slightly with temp.



- in SI units, the specific heat capacity of water is **4186 J / kg K**.
- **Mechanical Equivalent of Heat** -amount of work needed to produce 1 cal of heat
- Specific heat capacity depends on the process or the conditions under which capacity transfer takes place.
- Relation b/w **specific heat capacity at constant volume and specific heat capacity at constant pressure**.

$$C_p - C_v = R$$

❖ Thermodynamic state variables and equation of state-

- A gas allowed to expand freely against vacuum is not an equilibrium state.
- Thermodynamic state variables are of two type –
- 1) **Extensive** -indicate 'size' of the system.
e.g. **internal energy, volume and mass**
- 2) **Intensive** -do not depend on size or quantity.
e.g. **pressure, temperature and density**
- **ΔQ is not a state variable however it is proportional to the mass, and hence is extensive.**
- The product of an intensive variable like P and extensive like ΔV is 'extensive'.

❖ Thermodynamic processes –

1) **Quasistatic process** -such a process is infinitely slow, nearly static.

It remains in thermal and mechanical equilibrium with its surroundings throughout.

In this process, at every stage, the difference in pressure of system and external pressure is infinitesimally small.

2) **Isothermal process-** **Temp.(T) = fixed**

So by ideal gas equation, **PV = constant**

- Work done, **$W = nRT \ln(V_2/V_1)$**
- **For an ideal gas, internal energy depends only on temp, thus there is no change in the internal energy of an ideal gas in an Isothermal Process.**
- The first law implies that,

$$\Delta Q = \Delta W$$

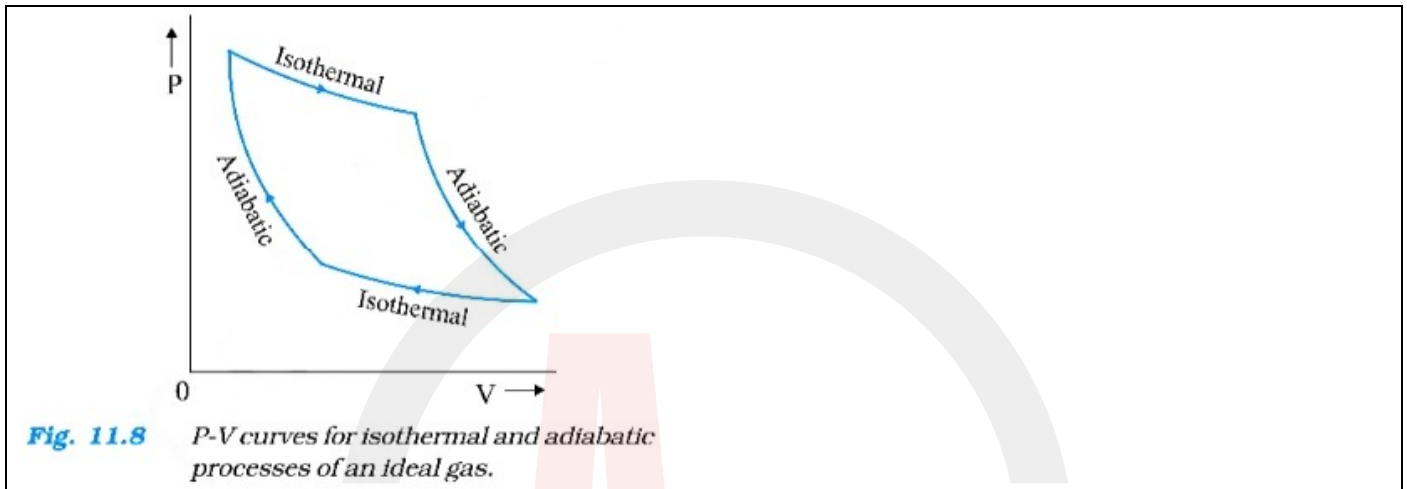
3) **Adiabatic process-** **heat absorbed or released is zero ($\Delta Q = 0$)**

- work done by gas results in decrease in its internal energy (hence its temp.)
- **$PV^\gamma = \text{constant}$**

Here, γ = ratio of specific heats

$$\gamma = C_p / C_v$$

- Work done, $W = nR(T_2 - T_1) / \gamma - 1$
- Work done by the gas, $W > 0$ then, $T_2 < T_1$
Work done on the gas, $W < 0$ then, $T_2 > T_1$



4) Isochoric process - $V = \text{constant}$

- No work done on or by the gas.
- Heat absorbed by gas goes entirely to change its internal energy and its temperature.

5) Isobaric process - $P = \text{constant}$

- Work done, $W = P(V_2 - V_1) = nR(T_2 - T_1)$
- The heat absorbed goes partly to increase internal energy and partly to do work.

6) Cyclic process -

- If system returns to its initial state .
- Internal energy, a state variable $\Delta U = 0$ for a cyclic process.
- Total heat absorbed equal the work done by the system .

❖ Second law of Thermodynamics-

➤ This law disallows some processes consistent with the first law of thermodynamics.

✓ Kelvin Plank statement –

“no process is possible whose sole result is the absorption of heat from a reservoir and the complete conversion of the heat into work”.

✓ Clausius statement-

“no process is possible whose sole result is the transfer of heat from a colder object to a hotter body”.

❖ Reversible and Irreversible Processes-

- A process is reversible if it can be reversed such that both system and surroundings return to their original states.
- **Spontaneous processes of nature are 'irreversible'.**
- The idealised reversible process is a quasi static process with no dissipative factors such as friction , viscosity , etc.

❖ Carnot Engine-

- It is the only reversible engine possible that works b/w two reservoirs at diff temperatures (T_1 is the temp. of source and T_2 is the temp. of sink)
- Efficiency of Carnot Engine ,
$$\eta = 1 - (T_2/T_1)$$
- No engine operating b/w two temperatures can have efficiency greater than that of carnot engine.
- In a carnot cycle,

$$Q_1/Q_2 = T_1/T_2$$

It is a universal relation independent of the nature of system.

SUMMARY

1. The zeroth law of thermodynamics states that '*two systems in thermal equilibrium with a third system separately are in thermal equilibrium with each other*'. The Zeroth Law leads to the concept of temperature.
2. Internal energy of a system is the sum of kinetic energies and potential energies of the molecular constituents of the system. It does not include the over-all kinetic energy of the system. Heat and work are two modes of energy transfer to the system. Heat is the energy transfer arising due to temperature difference between the system and the surroundings. Work is energy transfer brought about by other means, such as moving the piston of a cylinder containing the gas, by raising or lowering some weight connected to it.

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3. The first law of thermodynamics is the general law of conservation of energy applied to any system in which energy transfer from or to the surroundings (through heat and work) is taken into account. It states that

$$\Delta Q = \Delta U + \Delta W$$

where ΔQ is the heat supplied to the system, ΔW is the work done by the system and ΔU is the change in internal energy of the system.

4. The specific heat capacity of a substance is defined by

$$s = \frac{1}{m} \frac{\Delta Q}{\Delta T}$$

where m is the mass of the substance and ΔQ is the heat required to change its temperature by ΔT . The molar specific heat capacity of a substance is defined by

$$C = \frac{1}{\mu} \frac{\Delta Q}{\Delta T}$$

where μ is the number of moles of the substance. For a solid, the law of equipartition of energy gives

$$C = 3R$$

which generally agrees with experiment at ordinary temperatures.

Calorie is the old unit of heat. 1 calorie is the amount of heat required to raise the temperature of 1 g of water from 14.5 °C to 15.5 °C. 1 cal = 4.186 J.

5. For an ideal gas, the molar specific heat capacities at constant pressure and volume satisfy the relation

$$C_p - C_v = R$$

where R is the universal gas constant.

6. Equilibrium states of a thermodynamic system are described by state variables. The value of a state variable depends only on the particular state, not on the path used to arrive at that state. Examples of state variables are pressure (P), volume (V), temperature (T), and mass (m). Heat and work are not state variables. An Equation of State (like the ideal gas equation $PV = \mu RT$) is a relation connecting different state variables.
7. A quasi-static process is an infinitely slow process such that the system remains in thermal and mechanical equilibrium with the surroundings throughout. In a quasi-static process, the pressure and temperature of the environment can differ from those of the system only infinitesimally.
8. In an isothermal expansion of an ideal gas from volume V_1 to V_2 at temperature T the heat absorbed (Q) equals the work done (W) by the gas, each given by

$$Q = W = \mu R T \ln \left(\frac{V_2}{V_1} \right)$$

9. In an adiabatic process of an ideal gas

$$PV^\gamma = \text{constant}$$

where

$$\gamma = \frac{C_p}{C_v}$$

Work done by an ideal gas in an adiabatic change of state from (P_1, V_1, T_1) to (P_2, V_2, T_2) is

$$W = \frac{\mu R (T_1 - T_2)}{\gamma - 1}$$

10. The second law of thermodynamics disallows some processes consistent with the First Law of Thermodynamics. It states

Kelvin-Planck statement

No process is possible whose sole result is the absorption of heat from a reservoir and complete conversion of the heat into work.

Clausius statement

No process is possible whose sole result is the transfer of heat from a colder object to a hotter object.

Put simply, the Second Law implies that no heat engine can have efficiency η equal to 1 or no refrigerator can have co-efficient of performance α equal to infinity.

11. A process is reversible if it can be reversed such that both the system and the surroundings return to their original states, with no other change anywhere else in the universe. Spontaneous processes of nature are irreversible. The idealised reversible process is a quasi-static process with no dissipative factors such as friction, viscosity, etc.
12. Carnot engine is a reversible engine operating between two temperatures T_1 (source) and T_2 (sink). The Carnot cycle consists of two isothermal processes connected by two adiabatic processes. The efficiency of a Carnot engine is given by

$$\eta = 1 - \frac{T_2}{T_1} \quad (\text{Carnot engine})$$

No engine operating between two temperatures can have efficiency greater than that of the Carnot engine.

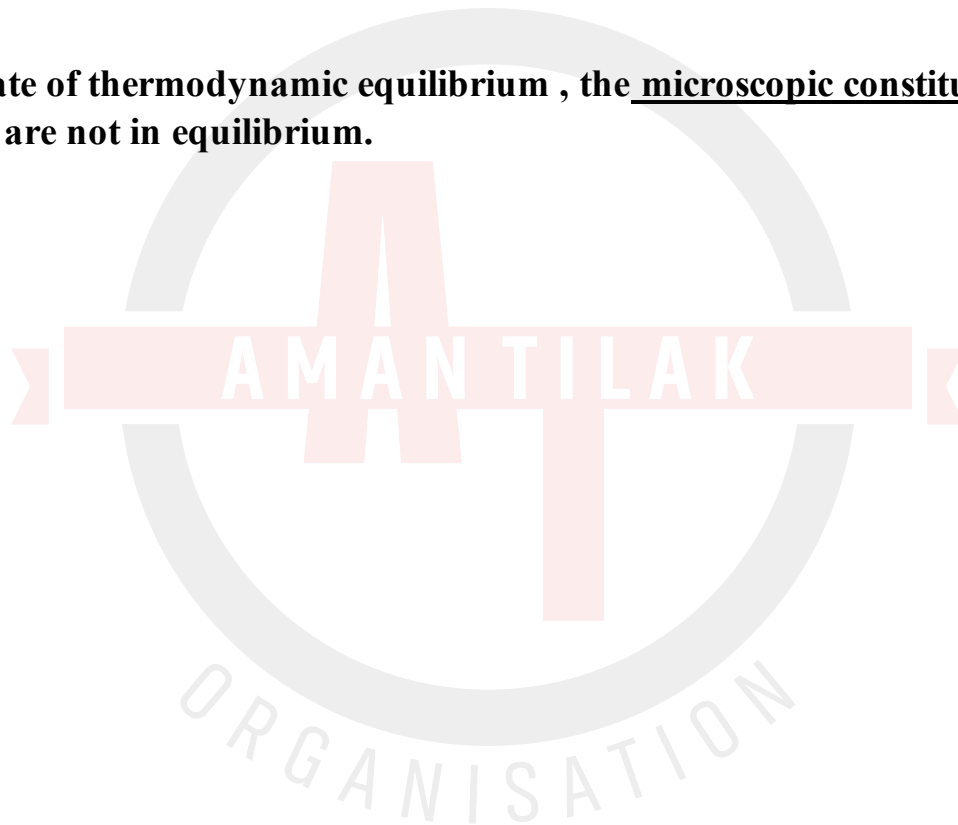
13. If $Q > 0$, heat is added to the system
If $Q < 0$, heat is removed to the system
If $W > 0$, Work is done by the system
If $W < 0$, Work is done on the system

POINTS TO PONDER

1. Temperature of a body is related to its average internal energy, not to the kinetic energy of motion of its centre of mass. A bullet fired from a gun is not at a higher temperature because of its high speed.
2. Equilibrium in thermodynamics refers to the situation when macroscopic variables describing the thermodynamic state of a system do not depend on time. Equilibrium of a system in mechanics means the net external force and torque on the system are zero.
3. In a state of thermodynamic equilibrium, the microscopic constituents of a system are not in equilibrium (in the sense of mechanics).
4. Heat capacity, in general, depends on the process the system goes through when heat is supplied.
5. In isothermal quasi-static processes, heat is absorbed or given out by the system even though at every stage the gas has the same temperature as that of the surrounding reservoir. This is possible because of the infinitesimal difference in temperature between the system and the reservoir.

✚ Importants from NCERT summary and Points to Ponder-

- If work is done by the system, $W > 0$
If work is done on the system, $W < 0$
- Temp of a body is related to its average internal energy ,not to kinetic energy of motion of COM. A bullet fired from a gun is not at a higher temp. because of its high speed.
- In a state of thermodynamic equilibrium , the microscopic constituents of a system are not in equilibrium.



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